```
> restart;#SchwarAx;
> with(PDEtools):
> with(Physics):
> with(LinearAlgebra):
> Setup(mathematicalnotation = true)
```

$$[mathematicalnotation = true] \tag{1}$$

```
> Setup(coordinatesystems = cartesian)
```

*Systems of spacetime coordinates are:* $\{X = (x, y, z, t)\}$

$$[coordinatesystems = \{X\}] \tag{2}$$

```
> g_[ ];
```

$$g_{\mu,\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{3}$$

```
> ds2 := r^2 dtheta^2 + r^2 sin(θ)^2 dphi^2 - r·c^2·dr^2/(2 m G - r·c^2) - (r c^2 - 2 m G)/r · dt^2 ; # The Schwarzchild metric
```

$$ds2 := r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - \frac{r\, c^2\, dr^2}{-r\, c^2 + 2\, m\, G} - \frac{(r\, c^2 - 2\, m\, G)\, dt^2}{r} \tag{4}$$

```
> Setup(coordinates = spherical, metric = ds2);
```

*Systems of spacetime coordinates are:* $\{X = (r, \theta, \phi, t)\}$

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*Coordinates:* $[r, \theta, \phi, t]$ *. Signature:* $(- - - +)$

---

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{r\, c^2}{-r\, c^2 + 2\, G\, m} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r\, c^2 + 2\, G\, m}{r} \end{bmatrix}$$

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*Setting lowercaselatin_is letters to represent space indices*

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$$\left[coordinatesystems = \{X\}, metric = \left\{(1,1) = -\frac{r\, c^2}{-r\, c^2 + 2\, G\, m}, (2,2) = r^2, (3,3) = r^2 \sin(\theta)^2, (4,4) = -\frac{r\, c^2 - 2\, G\, m}{r}\right\}, spaceindices = lowercaselatin_is\right] \tag{5}$$

```
> Ricci[scalarsdefinition];
```

$$\Phi_{00} = -\frac{R^{\mu,\nu}\, l_\mu\, l_\nu}{2}, \Phi_{01} = -\frac{R^{\mu,\nu}\, l_\mu\, m_\nu}{2}, \Phi_{02} = -\frac{R^{\mu,\nu}\, m_\mu\, m_\nu}{2}, \Phi_{11} = -\frac{R^{\mu,\nu}\, (l_\mu\, n_\nu + m_\mu\, \overline{m}_\nu)}{4}, \Phi_{12} = -\frac{R^{\mu,\nu}\, m_\mu\, n_\nu}{2}, \Phi_{22} = -\frac{R^{\mu,\nu}\, n_\mu\, n_\nu}{2}, \Lambda = \frac{R^{\mu}_{\ \mu}}{24} \tag{6}$$

> *Ricci*[*scalars*];

$$\Phi_{00}=0,\ \Phi_{01}=0,\ \Phi_{02}=0,\ \Phi_{11}=0,\ \Phi_{12}=0,\ \Phi_{22}=0,\ \Lambda=0 \tag{7}$$

> *Ricci*[*scalar*];

$$0 \tag{8}$$

> $TR := r = \frac{m\,G\,(1-\rho)}{c^2}$ # $Rs = \frac{2\,G\cdot m}{c^2}$ *the Schwarschields radius so we have* $r = \frac{Rs}{2}(1-\rho)$;

*The inverse function of* $\rho = 1 - \frac{2\cdot r}{Rs}$

$$TR := r = \frac{m\,G\,(1-\rho)}{c^2} \tag{9}$$

> *subs*(*r* = rho, *TransformCoordinates*(*TR*, *g_*[mu, nu], [rho, theta, phi, *t*]));#,*setmetric*

$$\begin{bmatrix} \frac{m^2 G^2 (-1+\rho)}{c^4 (1+\rho)} & 0 & 0 & 0 \\ 0 & \frac{m^2 G^2 (-1+\rho)^2}{c^4} & 0 & 0 \\ 0 & 0 & \frac{m^2 G^2 (-1+\rho)^2 \sin(\theta)^2}{c^4} & 0 \\ 0 & 0 & 0 & -\frac{(1+\rho)\,c^2}{-1+\rho} \end{bmatrix} \tag{10}$$

> *TransformCoordinates*(*TR*, *g_*[mu, nu], [rho, theta, phi, *t*], *output* = *line_element*)

$$\frac{G^2 m^2 (-1+\rho)\left(-\sin(\theta)^2 (\mathrm{d}(\phi))^2 + (\mathrm{d}(\rho))^2 - (\mathrm{d}(\theta))^2 + \left(\sin(\theta)^2 (\mathrm{d}(\phi))^2 + (\mathrm{d}(\theta))^2\right)\rho^2\right)}{c^4 (1+\rho)} - \frac{(1+\rho)\,c^2\,(\mathrm{d}(t))^2}{-1+\rho} \tag{11}$$

> *subs*(ρ = *r*, *TransformCoordinates*(*TR*, *g_*[mu, nu], [rho, theta, phi, *t*], *output* = *line_element*));

$$\frac{G^2 m^2 (-1+r)\left(-\sin(\theta)^2 (\mathrm{d}(\phi))^2 + (\mathrm{d}(r))^2 - (\mathrm{d}(\theta))^2 + \left(\sin(\theta)^2 (\mathrm{d}(\phi))^2 + (\mathrm{d}(\theta))^2\right) r^2\right)}{c^4 (1+r)} - \frac{(1+r)\,c^2\,(\mathrm{d}(t))^2}{-1+r} \tag{12}$$

> $ds21 := \frac{m^2 G^2 (-1+r)^2}{c^4}\cdot dtheta^2 + \frac{m^2 G^2 (-1+r)^2}{c^4}\cdot \sin(\theta)^2\, dphi^2 + \frac{m^2 G^2 (-1+r)}{c^4 (1+r)}\cdot dr^2 - \frac{(1+r)\,c^2}{-1+r}\cdot dt^2$;

$$ds21 := \frac{m^2 G^2 (-1+r)^2\, d\theta^2}{c^4} + \frac{m^2 G^2 (-1+r)^2 \sin(\theta)^2\, d\phi^2}{c^4} + \frac{m^2 G^2 (-1+r)\, dr^2}{c^4 (1+r)} - \frac{(1+r)\,c^2\,dt^2}{-1+r} \tag{13}$$

> *Setup*(*coordinates* = *spherical*, *metric* = *ds21*);

*Systems of spacetime coordinates are:* $\{X = (r, \theta, \phi, t)\}$

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*Coordinates:* $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

---

$$g_{\mu,\nu} = \begin{bmatrix} \frac{m^2 G^2 (-1+r)}{c^4 (1+r)} & 0 & 0 & 0 \\ 0 & \frac{m^2 G^2 (-1+r)^2}{c^4} & 0 & 0 \\ 0 & 0 & \frac{m^2 G^2 (-1+r)^2 \sin(\theta)^2}{c^4} & 0 \\ 0 & 0 & 0 & -\frac{(1+r) c^2}{-1+r} \end{bmatrix}$$

---

$$\left[ coordinatesystems = \{X\}, metric = \left\{ (1,1) = \frac{m^2 G^2 (-1+r)}{c^4 (1+r)}, (2,2) = \frac{m^2 G^2 (-1+r)^2}{c^4}, (3,3) = \frac{m^2 G^2 (-1+r)^2 \sin(\theta)^2}{c^4}, (4,4) = -\frac{(1+r) c^2}{-1+r} \right\} \right] \quad (14)$$

> $Ricci[scalarsdefinition];$

$$\Phi_{00} = -\frac{R^{\mu,\nu} l_\mu l_\nu}{2}, \Phi_{01} = -\frac{R^{\mu,\nu} l_\mu m_\nu}{2}, \Phi_{02} = -\frac{R^{\mu,\nu} m_\mu m_\nu}{2}, \Phi_{11} = -\frac{R^{\mu,\nu} \left(l_\mu n_\nu + m_\mu \overline{m}_\nu\right)}{4}, \Phi_{12} = -\frac{R^{\mu,\nu} m_\mu n_\nu}{2}, \Phi_{22} = -\frac{R^{\mu,\nu} n_\mu n_\nu}{2}, \Lambda = \frac{R^\mu}{24} \quad (15)$$

> $Ricci[scalars];$

$$\Phi_{00} = 0, \Phi_{01} = 0, \Phi_{02} = 0, \Phi_{11} = 0, \Phi_{12} = 0, \Phi_{22} = 0, \Lambda = 0 \quad (16)$$

> $Ricci[scalar];$

$$0 \quad (17)$$

> $EnergyMomentum[definition]; EnergyMomentum[\,]$

$$T_{\mu,\nu} = \frac{G_{\mu,\nu}}{8\pi}$$

$$T_{\mu,\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (18)$$

> $f(r,t) := \frac{m^2 G^2 (-1+r)^2}{c^4} \cdot dtheta^2 + \frac{m^2 G^2 (-1+r)^2}{c^4} \cdot \sin(\theta)^2 dphi^2 + \frac{m^2 G^2 (-1+r)}{c^4 (1+r)} \cdot dr^2 - \frac{(1+r) c^2}{-1+r} \cdot dt^2;$

$$f := (r,t) \mapsto m^2 G^2 (-1+r)^2 \frac{1}{c^4} d\theta^2 + m^2 G^2 (-1+r)^2 \frac{1}{c^4} \sin(\theta)^2 d\phi^2 + m^2 G^2 (-1+r) \frac{1}{c^4 (1+r)} dr^2 - (1+r) c^2 \frac{1}{-1+r} dt^2 \quad (19)$$

> $change_of_vars := \left\{ \rho = \frac{-r \cdot c^2 + m \cdot G}{m \cdot G} \right\}; \# \rho = 1 - \frac{2 \cdot r}{R_{Schwar}}; Invers\ Trans\ \rho = Rsch(1-r)/2; RSchwar$

$$= \frac{2 \cdot m \cdot G}{c^2};$$

$$change_of_vars := \left\{ \rho = \frac{-r c^2 + G m}{G m} \right\} \tag{20}$$

> $tmp := dchange(change_of_vars, [f(\rho, t)], [r], params = m, G, c, simp_proc);$

$$tmp := \left[ \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + G m}{G m} \right)^2 d\theta^2}{c^4} + \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + G m}{G m} \right)^2 \sin(\theta)^2 d\phi^2}{c^4} \right. \tag{21}$$

$$\left. + \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + G m}{G m} \right) dr^2}{c^4 \left( 1 + \frac{-r c^2 + G m}{G m} \right)} - \frac{\left( 1 + \frac{-r c^2 + G m}{G m} \right) c^2 dt^2}{-1 + \frac{-r c^2 + G m}{G m}} \right]$$

> $ds22 := \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + m G}{m G} \right)^2}{c^4} \cdot dtheta^2 + \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + m G}{m G} \right)^2}{c^4} \cdot \sin(\theta)^2 \cdot dphi^2$

$$+ \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + m G}{m G} \right)}{c^4 \left( 1 + \frac{-r c^2 + m G}{m G} \right)} \cdot dr^2 - \frac{\left( 1 + \frac{-r c^2 + m G}{m G} \right) c^2}{-1 + \frac{-r c^2 + m G}{m G}} \cdot dt^2;$$

$$ds22 := \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + G m}{G m} \right)^2 d\theta^2}{c^4} + \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + G m}{G m} \right)^2 \sin(\theta)^2 d\phi^2}{c^4} \tag{22}$$

$$+ \frac{m^2 G^2 \left( -1 + \frac{-r c^2 + G m}{G m} \right) dr^2}{c^4 \left( 1 + \frac{-r c^2 + G m}{G m} \right)} - \frac{\left( 1 + \frac{-r c^2 + G m}{G m} \right) c^2 dt^2}{-1 + \frac{-r c^2 + G m}{G m}}$$

> $Setup(coordinates = spherical, metric = ds22);$

*Systems of spacetime coordinates are:* $\{X = (r, \theta, \phi, t)\}$

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*Coordinates:* $[r, \theta, \phi, t]$. *Signature:* $(- - - +)$

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$$g_{\mu, \nu} = \begin{bmatrix} -\frac{m^2 G^2 r}{c^2 (-r c^2 + 2 G m)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r c^2 + 2 G m}{r} \end{bmatrix}$$

---

---

$$\left[ coordinatesystems = \{X\}, metric = \left\{ (1, 1) = -\frac{m^2 G^2 r}{c^2 (-r c^2 + 2 G m)}, (2, 2) = r^2, (3, 3) = r^2 \sin(\theta)^2, (4, 4) \right. \right. \tag{23}$$

$$= \frac{-r c^2 + 2 G m}{r} \Bigg\} \Bigg]$$

> *Ricci*[*scalarsdefinition*];

$$\Phi_{00} = -\frac{R^{\mu,\nu}\, l_\mu l_\nu}{2}, \Phi_{01} = -\frac{R^{\mu,\nu}\, l_\mu m_\nu}{2}, \Phi_{02} = -\frac{R^{\mu,\nu}\, m_\mu m_\nu}{2}, \Phi_{11} = -\frac{R^{\mu,\nu}\, \left(l_\mu n_\nu + m_\mu \overline{m}_\nu\right)}{4}, \Phi_{12} = -\frac{R^{\mu,\nu}\, m_\mu n_\nu}{2}, \Phi_{22} = -\frac{R^{\mu,\nu}\, n_\mu n_\nu}{2}, \Lambda = \frac{R^{\mu}_{\ \mu}}{24} \quad (24)$$

> *Ricci*[*scalars*];

$$\Phi_{00} = 0, \Phi_{01} = 0, \Phi_{02} = 0, \Phi_{11} = \frac{m^2 G^2 - c^4}{4 r^2 m^2 G^2}, \Phi_{12} = 0, \Phi_{22} = 0, \Lambda = \frac{m^2 G^2 - c^4}{12 r^2 m^2 G^2} \quad (25)$$

> *Ricci*[*scalar*];

$$\frac{2\left(m^2 G^2 - c^4\right)}{r^2 m^2 G^2} \quad (26)$$

> $g_[\ ]; \Lambda := \frac{m^2 G^2 - c^4}{12 r^2 m^2 G^2}$; *Ricci*[~μ, ν, *matrix*];

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{m^2 G^2 r}{c^2\left(-r c^2 + 2 G m\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-r c^2 + 2 G m}{r} \end{bmatrix}$$

$$\Lambda := \frac{m^2 G^2 - c^4}{12 r^2 m^2 G^2}$$

$$R^{\mu}_{\ \nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{m^2 G^2 - c^4}{r^2 m^2 G^2} & 0 & 0 \\ 0 & 0 & \frac{m^2 G^2 - c^4}{r^2 m^2 G^2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (27)$$

> $ds22 := r^2 \cdot dtheta^2 + r^2 \cdot \sin(\theta)^2 \cdot dphi^2 - \frac{m^2 G^2 r}{c^2\left(-r c^2 + 2 m G\right)} \cdot dr^2 + \frac{-r c^2 + 2 m G}{r} \cdot dt^2$;

*# We simplify the metric (22)*

$$ds22 := r^2 d\theta^2 + r^2 \sin(\theta)^2 d\phi^2 - \frac{m^2 G^2 r\, dr^2}{c^2\left(-r c^2 + 2 G m\right)} + \frac{\left(-r c^2 + 2 G m\right) dt^2}{r} \quad (28)$$

> $\left(\frac{1}{2} \cdot g_[\ ] \cdot Ricci[scalar] - Ricci[\sim\mu, \nu, matrix] + \Lambda \cdot g_[\ ]\right)$;

$$\frac{13\, g_{\mu,\nu}\left(m^2 G^2 - c^4\right)}{12 r^2 m^2 G^2} - R^{\mu}_{\ \nu} = \left[\left[-\frac{13\left(m^2 G^2 - c^4\right)}{12 r c^2\left(-r c^2 + 2 G m\right)}, 0, 0, 0\right], \left[0, \frac{13\left(m^2 G^2 - c^4\right)}{12 m^2 G^2} - \frac{m^2 G^2 - c^4}{r^2 m^2 G^2}, 0, 0\right], \right. \quad (29)$$

$$\left[0,\ 0,\ \frac{13\left(m^2G^2-c^4\right)\sin(\theta)^2}{12\,m^2G^2}-\frac{m^2G^2-c^4}{r^2m^2G^2},\ 0\right],$$
$$\left[0,\ 0,\ 0,\ \frac{13\left(m^2G^2-c^4\right)\left(-rc^2+2Gm\right)}{12\,r^3m^2G^2}\right]\Bigg]$$

> $$\frac{1}{2}\cdot\frac{2\left(m^2G^2-c^4\right)}{r^2m^2G^2}\cdot\begin{bmatrix}-\frac{m^2G^2r}{c^2\left(-rc^2+2mG\right)} & 0 & 0 & 0\\ 0 & r^2 & 0 & 0\\ 0 & 0 & r^2\sin(\theta)^2 & 0\\ 0 & 0 & 0 & \frac{-rc^2+2mG}{r}\end{bmatrix}$$
$$-\begin{bmatrix}0 & 0 & 0 & 0\\ 0 & \frac{m^2G^2-c^4}{r^2m^2G^2} & 0 & 0\\ 0 & 0 & \frac{m^2G^2-c^4}{r^2m^2G^2}\cdot\sin(\theta)^2 & 0\\ 0 & 0 & 0 & 0\end{bmatrix}+\frac{m^2G^2-c^4}{12\,r^2m^2G^2}$$
$$\cdot\begin{bmatrix}-\frac{m^2G^2r}{c^2\left(-rc^2+2mG\right)} & 0 & 0 & 0\\ 0 & r^2 & 0 & 0\\ 0 & 0 & r^2\sin(\theta)^2 & 0\\ 0 & 0 & 0 & \frac{-rc^2+2mG}{r}\end{bmatrix};$$

$$\left[\left[-\frac{13\left(m^2G^2-c^4\right)}{12\,rc^2\left(-rc^2+2Gm\right)},\ 0,\ 0,\ 0\right],\right. \tag{30}$$
$$\left[0,\ \frac{13\left(m^2G^2-c^4\right)}{12\,m^2G^2}-\frac{m^2G^2-c^4}{r^2m^2G^2},\ 0,\ 0\right],$$
$$\left[0,\ 0,\ \frac{13\left(m^2G^2-c^4\right)\sin(\theta)^2}{12\,m^2G^2}-\frac{\left(m^2G^2-c^4\right)\sin(\theta)^2}{r^2m^2G^2},\ 0\right],$$
$$\left.\left[0,\ 0,\ 0,\ \frac{13\left(m^2G^2-c^4\right)\left(-rc^2+2Gm\right)}{12\,r^3m^2G^2}\right]\right]$$

> $$factor\left(-\frac{13\left(m^2G^2-c^4\right)}{12\,rc^2\left(-rc^2+2mG\right)}\right);factor\left(\frac{13\left(m^2G^2-c^4\right)}{12\,m^2G^2}-\frac{m^2G^2-c^4}{r^2m^2G^2}\right);$$
$$factor\left(\frac{13\left(m^2G^2-c^4\right)\left(-rc^2+2mG\right)}{12\,r^3m^2G^2}\right);$$
$$-\frac{13\left(Gm-c^2\right)\left(Gm+c^2\right)}{12\,rc^2\left(-rc^2+2Gm\right)}$$
$$\frac{\left(13r^2-12\right)\left(Gm-c^2\right)\left(Gm+c^2\right)}{12\,r^2m^2G^2}$$

(31)

$$\frac{13\left(Gm-c^2\right)\left(Gm+c^2\right)\left(-rc^2+2Gm\right)}{12r^3m^2G^2} \tag{31}$$

>

```
> N := (m*G - c^2)*(m*G + c^2)
      .[[[-13/12, 1/(r*c^2*(-r*c^2 + 2*m*G)), 0, 0], [0, (13*r^2 - 12)/(12*r^2*m^2*G^2), 0, 0],
      [0, 0, (13*r^2 - 12)/(12*r^2*m^2*G^2)*sin(theta)^2, 0], [0, 0, 0, 13*(-r*c^2 + 2*m*G)/(12*r^3*m^2*G^2)]]];
```

$$N := \begin{bmatrix} -\frac{13\left(Gm-c^2\right)\left(Gm+c^2\right)}{12rc^2\left(-rc^2+2Gm\right)} & 0 & 0 & 0 \\ 0 & \frac{\left(13r^2-12\right)\left(Gm-c^2\right)\left(Gm+c^2\right)}{12r^2m^2G^2} & 0 & 0 \\ 0 & 0 & \frac{\left(Gm-c^2\right)\left(Gm+c^2\right)\left(13r^2-12\right)\sin(\theta)^2}{12r^2m^2G^2} & 0 \\ 0 & 0 & 0 & \frac{\left(Gm-c^2\right)\left(Gm+c^2\right)\left(-13rc^2+26Gm\right)}{12r^3m^2G^2} \end{bmatrix} \tag{32}$$

```
> solve(-13*(m*G - c^2)*(m*G + c^2)/(12*r*c^2*(-r*c^2 + 2*m*G)) = 0);
```

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{c^2}{m}, c=c, m=m, r=r\right\} \tag{33}$$

```
> solve((13*r^2 - 12)*(m*G - c^2)*(m*G + c^2)/(12*r^2*m^2*G^2) = 0);
```

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=G, c=c, m=m, r=\frac{2\sqrt{39}}{13}\right\}, \left\{G=G, c=c, m=m, r=-\frac{2\sqrt{39}}{13}\right\}, \left\{G=\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{c^2}{m}, c=c, m=m, r=r\right\} \tag{34}$$

```
> solve(13*(m*G - c^2)*(m*G + c^2)*(-r*c^2 + 2*m*G)/(12*r^3*m^2*G^2));
```

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=-\frac{c^2}{m}, c=c, m=m, r=r\right\}, \left\{G=\frac{rc^2}{2m}, c=c, m=m, r=r\right\} \tag{35}$$

```
> N := Matrix(4, 4, [[0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0], [0, 0, 0, 0]]);#For m=c^2/G we have null matrix thus null Tensor StresEnergy
```

$$N := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \text{;\#For } m=\frac{c^2}{G} \text{ we have null matrix thus null Tensor StresEnergy}$$

$$N := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{36}$$

> $EnergyMomentum_{\mu,\tau} = \left( \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \right)$

$$T_{\mu,\tau} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \tag{37}$$

> $restart;$

> $with(Physics):$

> $with(PDEtools):$

> $with(LinearAlgebra):$

> $Setup(mathematicalnotation = true)$

$$[mathematicalnotation = true] \tag{38}$$

> $Setup(coordinatesystems = cartesian)$

$$\text{Systems of spacetime coordinates are:} \quad \{X = (x, y, z, t)\}$$

$$[coordinatesystems = \{X\}] \tag{39}$$

> $g_[\,];$

$$g_{\mu,\nu} = \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{40}$$

> $ds22 := r^2 \cdot dtheta^2 + r^2 \cdot \sin(\theta)^2 \cdot dphi^2 - \frac{m^2 G^2 r}{c^2 (-r c^2 + 2 m G)} \cdot dr^2 + \frac{-r c^2 + 2 m G}{r} \cdot dt^2;$

$$ds22 := r^2\, d\theta^2 + r^2 \sin(\theta)^2\, d\phi^2 - \frac{m^2 G^2 r\, dr^2}{c^2 (-r c^2 + 2 m G)} + \frac{(-r c^2 + 2 m G)\, dt^2}{r} \tag{41}$$

> $Setup(coordinates = spherical, metric = ds22);$

$$\text{Systems of spacetime coordinates are:} \quad \{X = (r, \theta, \phi, t)\}$$

---

$$\text{Coordinates: } [r, \theta, \phi, t]. \text{ Signature: } (- - - +)$$

---

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{m^2 G^2 r}{c^2 (-c^2 r + 2 G m)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2 \sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-c^2 r + 2 G m}{r} \end{bmatrix}$$

Setting lowercaselatin_is letters to represent space indices

$$\left[coordinatesystems = \{X\},\ metric = \left\{(1,1) = -\frac{m^2 G^2 r}{c^2\left(-c^2 r + 2 G m\right)},\ (2,2) = r^2,\ (3,3) = r^2 \sin(\theta)^2,\ (4,4) = \frac{-c^2 r + 2 G m}{r}\right\},\ spaceindices = lowercaselatin_is\right] \quad (42)$$

```
> simplify( - m^2*G^2*r/(c^2*(-r*c^2+2*m*G))*dr^2 ); simplify( (-r*c^2+2*m*G)/r*dt^2 );
```

$$-\frac{m^2 G^2 r\, dr^2}{c^2\left(-c^2 r + 2 G m\right)}$$

$$\frac{-c^2 dt^2 r + 2 G dt^2 m}{r} \quad (43)$$

```
> f(r,t,m) := r^2*dtheta^2 + r^2*sin(theta)^2*dphi^2 - m^2*G^2*r/(c^2*(-r*c^2+2*m*G))*dr^2 + (-r*c^2+2*m*G)/r*dt^2;
```

$$f := (r,t,m) \mapsto r^2 d\theta^2 + r^2 \sin(\theta)^2 d\phi^2 - m^2 G^2 r \frac{1}{c^2\left(-r c^2 + 2 m G\right)} dr^2 + \left(-r c^2 + 2 m G\right)\frac{1}{r} dt^2 \quad (44)$$

```
> change_of_vars := {m = 2*M/Rs};
  # The change of variable help us to cancel the Energy Tensor and prove the initial equation
```

$$change_of_vars := \left\{m = \frac{2 M}{Rs}\right\} \quad (45)$$

```
> tmp := dchange(change_of_vars, [f(r,t,m)], [M], r, t, m, );
```

$$tmp := \left[r^2 d\theta^2 + r^2 \sin(\theta)^2 d\phi^2 - \frac{4 M^2 G^2 r\, dr^2}{Rs^2 c^2\left(-c^2 r + \frac{4 G M}{Rs}\right)} + \frac{\left(-c^2 r + \frac{4 G M}{Rs}\right) dt^2}{r}\right] \quad (46)$$

```
> ds2 := r^2*dtheta^2 + r^2*sin(theta)^2*dphi^2 - 4*M^2*G^2*r*dr^2/(Rs^2*c^2*(-r*c^2+4*M*G/Rs)) + (-r*c^2+4*M*G/Rs)/r*dt^2;
```

$$ds2 := r^2 d\theta^2 + r^2 \sin(\theta)^2 d\phi^2 - \frac{4 M^2 G^2 r\, dr^2}{Rs^2 c^2\left(-c^2 r + \frac{4 G M}{Rs}\right)} + \frac{\left(-c^2 r + \frac{4 G M}{Rs}\right) dt^2}{r} \quad (47)$$

```
> Setup(coordinates = spherical, metric = ds2);
```

Systems of spacetime coordinates are: $\{X = (r, \theta, \phi, t)\}$

Coordinates: $[r, \theta, \phi, t]$. Signature: $(- - - +)$

$$g_{\mu,\nu} = \begin{bmatrix} -\frac{4M^2G^2r}{Rs\,c^2\left(-c^2 r Rs + 4GM\right)} & 0 & 0 & 0 \\ 0 & r^2 & 0 & 0 \\ 0 & 0 & r^2\sin(\theta)^2 & 0 \\ 0 & 0 & 0 & \frac{-c^2 r Rs + 4GM}{Rs\,r} \end{bmatrix}$$

---

$$\left[coordinatesystems = \{X\},\ metric = \left\{(1,1) = -\frac{4M^2G^2r}{Rs\,c^2\left(-c^2 r Rs + 4GM\right)},\ (2,2) = r^2,\ (3,3) = r^2\sin(\theta)^2,\ (4,4) = \frac{-c^2 r Rs + 4GM}{Rs\,r}\right\}\right] \quad (48)$$

> $Ricci[scalarsdefinition];$

$$\Phi_{00} = -\frac{R^{\mu,\nu}\,l_\mu\,l_\nu}{2},\ \Phi_{01} = -\frac{R^{\mu,\nu}\,l_\mu\,m_\nu}{2},\ \Phi_{02} = -\frac{R^{\mu,\nu}\,m_\mu\,m_\nu}{2},\ \Phi_{11} = -\frac{R^{\mu,\nu}\left(l_\mu\,n_\nu + m_\mu\,\overline{m}_\nu\right)}{4},\ \Phi_{12} = -\frac{R^{\mu,\nu}\,m_\mu\,n_\nu}{2},\ \Phi_{22} = -\frac{R^{\mu,\nu}\,n_\mu\,n_\nu}{2},\ \Lambda = \frac{R^{\mu}}{24} \quad (49)$$

> $Ricci[scalars];$

$$\Phi_{00} = 0,\ \Phi_{01} = 0,\ \Phi_{02} = 0,\ \Phi_{11} = \frac{-Rs^2c^4 + 4M^2G^2}{16r^2M^2G^2},\ \Phi_{12} = 0,\ \Phi_{22} = 0,\ \Lambda = \frac{-Rs^2c^4 + 4M^2G^2}{48r^2M^2G^2} \quad (50)$$

> $Ricci[scalar];$

$$\frac{-Rs^2c^4 + 4M^2G^2}{2r^2M^2G^2} \quad (51)$$

> $Ricci[scalar]$

$$\frac{-Rs^2c^4 + 4M^2G^2}{2r^2M^2G^2} \quad (52)$$

>

> $\Lambda := \frac{-Rs^2c^4 + 4M^2G^2}{48r^2M^2G^2};\ Ricci[\sim\mu, \nu, matrix];$

$$\Lambda := \frac{-Rs^2c^4 + 4M^2G^2}{48r^2M^2G^2}$$

$$R^{\mu}{}_{\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & \frac{-Rs^2c^4 + 4M^2G^2}{4r^2M^2G^2} & 0 & 0 \\ 0 & 0 & \frac{-Rs^2c^4 + 4M^2G^2}{4r^2M^2G^2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (53)$$

> $\frac{1}{2}\cdot g_[\ ]\cdot Ricci[scalar] - Ricci[\sim\mu, \nu, matrix] + \Lambda\cdot g_[\ ]$

*#Thus the equation for τ neutriono generated from the Sun we make the transformation* $m = \frac{2\cdot M}{Rs}$

$$\frac{13\, g_{\mu,\nu}\left(-Rs^2 c^4+4 M^2 G^2\right)}{48 r^2 M^2 G^2}-R^{\mu}{}_{\nu}=\left[\left[-\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)}{12 r Rs c^2\left(-c^2 r Rs+4 G M\right)},0,0,0\right],\right. \tag{54}$$

$$\left[0,\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)}{48 M^2 G^2}-\frac{-Rs^2 c^4+4 M^2 G^2}{4 r^2 M^2 G^2},0,0\right],$$

$$\left[0,0,\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)\sin(\theta)^2}{48 M^2 G^2}-\frac{-Rs^2 c^4+4 M^2 G^2}{4 r^2 M^2 G^2},0\right],$$

$$\left.\left[0,0,0,\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)\left(-c^2 r Rs+4 G M\right)}{48 r^3 M^2 G^2 Rs}\right]\right]$$

> $N:=\left[\left[-\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)}{12 r Rs c^2\left(-r c^2 Rs+4 M G\right)},0,0,0\right],\right.$
$\left[0,\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)}{48 M^2 G^2}-\frac{-Rs^2 c^4+4 M^2 G^2}{4 r^2 M^2 G^2},0,0\right],$
$\left[0,0,\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)\sin(\theta)^2}{48 M^2 G^2}-\frac{-Rs^2 c^4+4 M^2 G^2}{4 r^2 M^2 G^2}\cdot\sin(\theta)^2,0\right],$
$\left.\left[0,0,0,\frac{13\left(-Rs^2 c^4+4 M^2 G^2\right)\left(-r c^2 Rs+4 M G\right)}{48 r^3 M^2 G^2 Rs}\right]\right];$

$$N:=\left[\left[-\frac{-13 Rs^2 c^4+52 M^2 G^2}{12 c^2 r Rs\left(-c^2 r Rs+4 G M\right)},0,0,0\right],\right. \tag{55}$$

$$\left[0,\frac{-13 Rs^2 c^4+52 M^2 G^2}{48 M^2 G^2}-\frac{-Rs^2 c^4+4 M^2 G^2}{4 r^2 M^2 G^2},0,0\right],$$

$$\left[0,0,\frac{\left(-13 Rs^2 c^4+52 M^2 G^2\right)\sin(\theta)^2}{48 M^2 G^2}-\frac{\left(-Rs^2 c^4+4 M^2 G^2\right)\sin(\theta)^2}{4 r^2 M^2 G^2},0\right],$$

$$\left.\left[0,0,0,\frac{\left(-13 Rs^2 c^4+52 M^2 G^2\right)\left(-c^2 r Rs+4 G M\right)}{48 r^3 M^2 G^2 Rs}\right]\right]$$

> $a:=N[1,1];\ b:=N[4,4];\ d:=N[4,4];$

$$a:=-\frac{-13 Rs^2 c^4+52 M^2 G^2}{12 c^2 r Rs\left(-c^2 r Rs+4 G M\right)}$$

$$b:=\frac{\left(-13 Rs^2 c^4+52 M^2 G^2\right)\left(-c^2 r Rs+4 G M\right)}{48 r^3 M^2 G^2 Rs} \tag{56}$$

$$d:=\frac{\left(-13 Rs^2 c^4+52 M^2 G^2\right)\left(-c^2 r Rs+4 G M\right)}{48 r^3 M^2 G^2 Rs}$$

> $solve(a=0);$

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{Rs\, c^2}{2 M}, M=M, Rs=Rs, c=c, r=r\right\},\left\{G=-\frac{Rs\, c^2}{2 M}, M=M, Rs=Rs, c=c, r=r\right\} \tag{57}$$

> $solve(b=0);$

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{Rs\, c^2}{2 M}, M=M, Rs=Rs, c=c, r=r\right\},\left\{G=-\frac{Rs\, c^2}{2 M}, M=M, Rs=Rs, c=c, r=r\right\},\left\{G=\frac{c^2 r Rs}{4 M}, M=M,\right. \tag{58}$$

$$\left.Rs=Rs, c=c, r=r\right\}$$

> $solve(d=0)$

Warning, solve may be ignoring assumptions on the input variables.

$$\left\{G=\frac{Rs\,c^2}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=-\frac{Rs\,c^2}{2\,M}, M=M, Rs=Rs, c=c, r=r\right\}, \left\{G=\frac{c^2\,r\,Rs}{4\,M}, M=M, Rs=Rs, c=c, r=r\right\} \quad (59)$$

> $N0 := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$;#Thus for $Rs = G\cdot\frac{2\,M}{c^2}$;

$$N0 := \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (60)$$

> $EnergyMomentum[definition]$ ; $EnergyMomentum[\ ]$

$$T_{\mu,\nu} = \frac{G_{\mu,\nu}}{8\,\pi}$$

$$T_{\mu,\nu} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad (61)$$

> $\frac{1}{2}\cdot g_[\ ]\cdot Ricci[scalar] - Ricci[\sim\mu, \nu, matrix] + \Lambda\cdot g_[\ ] = k\cdot T_{\mu,\nu}$ # *Thus the equation is valid;*

$$\left(\frac{13\,g_{\mu,\nu}\left(-Rs^2\,c^4+4\,M^2\,G^2\right)}{48\,r^2\,M^2\,G^2} - R^{\mu}{}_{\nu} = \left[\left[-\frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)}{12\,r\,Rs\,c^2\left(-c^2\,r\,Rs+4\,G\,M\right)}, 0, 0, 0\right], \left[0, \frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)}{48\,M^2\,G^2} - \frac{-Rs^2\,c^4+4\,M^2\,G^2}{4\,r^2\,M^2\,G^2}, 0, 0\right], \left[0, 0, \frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)\sin(\theta)^2}{48\,M^2\,G^2} - \frac{-Rs^2\,c^4+4\,M^2\,G^2}{4\,r^2\,M^2\,G^2}, 0\right], \left[0, 0, 0, \frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)\left(-c^2\,r\,Rs+4\,G\,M\right)}{48\,r^3\,M^2\,G^2\,Rs}\right]\right) = k\,T_{\mu,\nu} \quad (62)$$

> $factor\left(-\frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)}{12\,r\,Rs\,c^2\left(-r\,c^2\,Rs+4\,M\,G\right)}\right)$; $factor\left(\frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)}{48\,M^2\,G^2} - \frac{-Rs^2\,c^4+4\,M^2\,G^2}{4\,r^2\,M^2\,G^2}\right)$; $factor\left(\frac{13\left(-Rs^2\,c^4+4\,M^2\,G^2\right)\left(-r\,c^2\,Rs+4\,M\,G\right)}{48\,r^3\,M^2\,G^2\,Rs}\right)$;

$$-\frac{13\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)}{12\,c^2\,r\,Rs\left(-c^2\,r\,Rs+4\,G\,M\right)}$$

$$\frac{\left(13\,r^2-12\right)\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)}{48\,r^2\,M^2\,G^2}$$

$$\frac{13\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)\left(-c^2\,r\,Rs+4\,G\,M\right)}{48\,r^3\,M^2\,G^2\,Rs} \quad (63)$$

```
> (-c^2 Rs + 2 M G) (c^2 Rs + 2 M G)
      .[[[ -13/(12 r Rs c^2 (-r c^2 Rs + 4 M G)), 0, 0, 0],
      [0, (13 r^2 - 12)/(48 M^2 G^2), 0, 0],
      [0, 0, (13 r^2 - 12)/(48 M^2 G^2) .sin(θ)^2, 0],
      [0, 0, 0, 13 .(-r c^2 Rs + 4 M G)/(48 r^3 M^2 G^2 Rs)]];
```

$$\left[\left[-\frac{13\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)}{12\,c^2\,r\,Rs\left(-c^2\,r\,Rs+4\,G\,M\right)},0,0,0\right],\left[0,\frac{\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)\left(13\,r^2-12\right)}{48\,M^2\,G^2},0,0\right],\left[0,0,\frac{\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)\left(13\,r^2-12\right)\sin(\theta)^2}{48\,M^2\,G^2},0\right],\left[0,0,0,\frac{\left(-Rs\,c^2+2\,G\,M\right)\left(Rs\,c^2+2\,G\,M\right)\left(-13\,c^2\,r\,Rs+52\,G\,M\right)}{48\,r^3\,M^2\,G^2\,Rs}\right]\right] \tag{64}$$

```
>
```

```
> series(-13/(12 r Rs c^2 (-r c^2 Rs + 4 M G)), r); series((13 r^2 - 12)/(48 M^2 G^2), r); series(13 .(-r c^2 Rs + 4 M G)/(48 r^3 M^2 G^2 Rs),
      r);
```

$$-\frac{13}{48\,c^2\,Rs\,G\,M}\,r^{-1}-\frac{13}{192}\,\frac{1}{M^2\,G^2}-\frac{13}{768}\,\frac{Rs\,c^2}{M^3\,G^3}\,r-\frac{13}{3072}\,\frac{Rs^2\,c^4}{M^4\,G^4}\,r^2-\frac{13}{12288}\,\frac{Rs^3\,c^6}{M^5\,G^5}\,r^3-\frac{13}{49152}\,\frac{Rs^4\,c^8}{M^6\,G^6}\,r^4-\frac{13}{196608}\,\frac{Rs^5\,c^{10}}{M^7\,G^7}\,r^5+O\left(r^6\right)$$

$$-\frac{1}{4\,M^2\,G^2}+\frac{13}{48}\,\frac{1}{M^2\,G^2}\,r^2$$

$$\frac{13}{12\,G\,M\,Rs}\,r^{-3}-\frac{13}{48}\,\frac{c^2}{M^2\,G^2}\,r^{-2} \tag{65}$$

```
> solve( - Rs. c^2 + 2. M.G);
```

$$\left\{G=\frac{Rs\,c^2}{2\,M},M=M,Rs=Rs,c=c\right\} \tag{66}$$

```
> c := 299792458; m := 1.9885.10^30; G := 6.6743.10^-11 ;
```

$$c := 299792458$$

$$m := 1.988500000 \times 10^{30}$$

$$G := 6.674300000 \times 10^{-11} \tag{67}$$

```
> Rs := evalf(2. m.G/c^2);
```

$$Rs := 2953.383938 \tag{68}$$

```
> M := c^2.Rs/(2. G);
```

$$M := 1.988500000 \times 10^{30} \quad \textbf{(69)}$$

> $-Rs \cdot c^2 + 2 \cdot M \cdot G;$

$$1. \times 10^{11} \quad \textbf{(70)}$$

>